

Differential Equations

ANSWER KEY



Solving by separation of variables

- 1 Solve the differential equation $\frac{dy}{dx} = \frac{x}{y}$, given that $y = 3$ when $x = 0$.
Separate: $y dy = x dx$. Integrate: $\frac{y^2}{2} = \frac{x^2}{2} + C$. Using $y = 3, x = 0$: $\frac{9}{2} = C$. So $y^2 = x^2 + 9$, giving $y = \sqrt{x^2 + 9}$ (taking the positive root since $y = 3 > 0$).
- 2 Solve $\frac{dy}{dx} = 2xy$, given that $y = 5$ when $x = 0$.
Separate: $\frac{dy}{y} = 2x dx$. Integrate: $\ln|y| = x^2 + C$. So $y = Ae^{x^2}$ for some constant A . Using $y = 5, x = 0$: $A = 5$. Solution: $y = 5e^{x^2}$.

Modeling with differential equations

- 3 A population P grows at a rate proportional to its current size, modeled by $\frac{dP}{dt} = kP$. If the population doubles every 5 years, find k , to three decimal places.
Solving the DE: $P(t) = P_0 e^{kt}$. Doubling: $2P_0 = P_0 e^{5k}$, so $e^{5k} = 2$, giving $5k = \ln 2$, $k = \frac{\ln 2}{5} \approx 0.139$.
- 4 Newton's law of cooling states $\frac{dT}{dt} = -k(T - T_a)$, where T_a is ambient temperature. Solve this differential equation for $T(t)$ in terms of the initial temperature T_0 .
Separate: $\frac{dT}{T - T_a} = -k dt$. Integrate: $\ln|T - T_a| = -kt + C$. So $T - T_a = Ae^{-kt}$. At $t = 0, T = T_0$, so $A = T_0 - T_a$. Solution: $T(t) = T_a + (T_0 - T_a)e^{-kt}$.

First-order linear differential equations

- 5 Solve $\frac{dy}{dx} + 2y = 6$ using an integrating factor, given $y = 1$ when $x = 0$.
Integrating factor: $e^{\int 2 dx} = e^{2x}$. Multiply through: $e^{2x} \frac{dy}{dx} + 2e^{2x}y = 6e^{2x}$, i.e. $\frac{d}{dx}(ye^{2x}) = 6e^{2x}$. Integrate: $ye^{2x} = 3e^{2x} + C$, so $y = 3 + Ce^{-2x}$. Using $y = 1, x = 0$: $1 = 3 + C$, so $C = -2$. Solution: $y = 3 - 2e^{-2x}$.

Euler's method for numerical solutions

- 6 Use Euler's method with step size $h = 0.5$ to estimate $y(1)$ for $\frac{dy}{dx} = x + y$, given $y(0) = 1$, showing each step.
Euler's method: $y_{n+1} = y_n + h \cdot f(x_n, y_n)$. Step 1 ($x_0 = 0, y_0 = 1$): $y_1 = 1 + 0.5(0 + 1) = 1.5$. Step 2 ($x_1 = 0.5, y_1 = 1.5$): $y_2 = 1.5 + 0.5(0.5 + 1.5) = 1.5 + 1 = 2.5$. Estimate: $y(1) \approx 2.5$.

Second-order context: simple harmonic motion

- 7 The differential equation $\frac{d^2x}{dt^2} = -\omega^2x$ describes simple harmonic motion. Verify that $x(t) = A\cos(\omega t)$ is a solution.
- $\frac{dx}{dt} = -A\omega\sin(\omega t)$. $\frac{d^2x}{dt^2} = -A\omega^2\cos(\omega t) = -\omega^2(A\cos(\omega t)) = -\omega^2x$, confirming $x(t) = A\cos(\omega t)$ satisfies the differential equation.

Family of solution curves and particular solutions

- 8 Find the general solution of $\frac{dy}{dx} = 3x^2$, and explain what the arbitrary constant represents geometrically.
- Integrating directly: $y = x^3 + C$. The constant C represents a vertical shift — the general solution is a whole family of curves, all with the same shape (since they share the same derivative $3x^2$ at every x), just shifted up or down by different amounts.
- 9 A tank drains such that $\frac{dV}{dt} = -k\sqrt{V}$ for some constant $k > 0$, where V is the volume of water. Solve this differential equation for $V(t)$, given $V(0) = V_0$.
- Separate: $\frac{dV}{\sqrt{V}} = -k dt$, i.e. $V^{-1/2} dV = -k dt$. Integrate: $2\sqrt{V} = -kt + C$. Using $V(0) = V_0$: $2\sqrt{V_0} = C$. So $2\sqrt{V} = -kt + 2\sqrt{V_0}$, giving $\sqrt{V} = \sqrt{V_0} - \frac{kt}{2}$, so $V(t) = \left(\sqrt{V_0} - \frac{kt}{2}\right)^2$.

Verifying a solution to a second-order differential equation

- 10 Verify that $y = e^{2x}$ is a solution of the differential equation $\frac{d^2y}{dx^2} - 5\frac{dy}{dx} + 6y = 0$.
- $\frac{dy}{dx} = 2e^{2x}$. $\frac{d^2y}{dx^2} = 4e^{2x}$. Substituting: $4e^{2x} - 5(2e^{2x}) + 6e^{2x} = 4e^{2x} - 10e^{2x} + 6e^{2x} = 0e^{2x} = 0$, confirming $y = e^{2x}$ is a solution.

Slope fields and qualitative solutions

- 11 A differential equation is given by $\frac{dy}{dx} = x - y$. Find the gradient of the solution curve at the points $(0, 0)$, $(1, 0)$, and $(1, 1)$, and explain what a slope field represents.
- At $(0, 0)$: gradient = $0 - 0 = 0$. At $(1, 0)$: gradient = $1 - 0 = 1$. At $(1, 1)$: gradient = $1 - 1 = 0$. A slope field is a diagram that plots a short line segment of the correct gradient at many points in the plane, without solving the equation explicitly — sketching a solution curve then amounts to drawing a curve that follows the direction of the segments at every point it passes through.

Solving a linear differential equation with an initial condition

- 12 Solve $\frac{dy}{dx} - \frac{y}{x} = x$ for $x > 0$ using an integrating factor, given $y = 2$ when $x = 1$.
- Rewrite in standard form: $\frac{dy}{dx} - \frac{1}{x}y = x$. Integrating factor: $e^{\int -\frac{1}{x}dx} = e^{-\ln x} = \frac{1}{x}$. Multiply through: $\frac{1}{x}\frac{dy}{dx} - \frac{1}{x^2}y = 1$, i.e. $\frac{d}{dx}\left(\frac{y}{x}\right) = 1$. Integrate: $\frac{y}{x} = x + C$, so $y = x^2 + Cx$. Using $y = 2, x = 1$: $2 = 1 + C$, so $C = 1$. Solution: $y = x^2 + x$.

A logistic growth model

- 13 A population P grows according to the logistic differential equation $\frac{dP}{dt} = kP(10 - P)$. Using the partial fraction decomposition $\frac{1}{P(10 - P)} = \frac{1}{10P} + \frac{1}{10(10 - P)}$, solve for P in terms of t , given $P(0) = 1$ and leaving k as an unknown constant.

Separate: $\frac{dP}{P(10 - P)} = k dt$. Integrate the left side using the given decomposition:
 $\frac{1}{10} \ln|P| - \frac{1}{10} \ln|10 - P| = kt + C$, i.e. $\frac{1}{10} \ln \left| \frac{P}{10 - P} \right| = kt + C$. Using $P(0) = 1$: $\frac{1}{10} \ln \left(\frac{1}{9} \right) = C$. So
 $\ln \left| \frac{P}{10 - P} \right| = 10kt + \ln \left(\frac{1}{9} \right)$, giving $\frac{P}{10 - P} = \frac{1}{9} e^{10kt}$. Solving for P : $9P = (10 - P)e^{10kt}$, so
 $P(9 + e^{10kt}) = 10e^{10kt}$, giving $P = \frac{10e^{10kt}}{9 + e^{10kt}}$.

Checking the order and degree of a differential equation

- 14 State the order of each of the following differential equations: (a) $\frac{dy}{dx} = 3x^2 - 2y$, (b) $\frac{d^2y}{dx^2} + 4\frac{dy}{dx} - y = 0$, (c) $\frac{d^3y}{dx^3} = \sin x$.
The order of a differential equation is the order of the highest derivative present. (a) Only a first derivative appears, so this is a first-order differential equation. (b) The highest derivative is $\frac{d^2y}{dx^2}$, so this is second-order. (c) The highest derivative is $\frac{d^3y}{dx^3}$, so this is third-order.

Using Euler's method with a smaller step size

- 15 Use Euler's method with step size $h = 0.25$ to estimate $y(0.5)$ for $\frac{dy}{dx} = 2x$, given $y(0) = 1$, showing each step, and compare with the exact solution $y = x^2 + 1$.
Euler's method: $y_{n+1} = y_n + h \cdot f(x_n, y_n)$. Step 1 ($x_0 = 0, y_0 = 1$): $y_1 = 1 + 0.25(2 \cdot 0) = 1 + 0 = 1$. Step 2 ($x_1 = 0.25, y_1 = 1$): $y_2 = 1 + 0.25(2 \cdot 0.25) = 1 + 0.125 = 1.125$. Estimate: $y(0.5) \approx 1.125$. Exact solution: $y(0.5) = (0.5)^2 + 1 = 1.25$. The Euler estimate underestimates the true value, since the method uses the derivative only at the start of each interval rather than accounting for how the derivative changes within it.
- 16 Solve the differential equation $\frac{dy}{dx} = \frac{y}{x}$ for $x > 0$, given $y = 6$ when $x = 2$, and describe the family of curves this represents.
Separate: $\frac{dy}{y} = \frac{dx}{x}$. Integrate: $\ln|y| = \ln|x| + C$, so $y = Ax$ for some constant A . Using $y = 6, x = 2$: $6 = 2A$, so $A = 3$. Solution: $y = 3x$. This represents the family of straight lines through the origin; the particular solution satisfying the initial condition is the specific line with slope 3.
- 17 A cup of coffee at 90°C is placed in a room at ambient temperature 20°C . Using Newton's law of cooling, $T(t) = T_a + (T_0 - T_a)e^{-kt}$, and given that after 10 minutes the coffee has cooled to 60°C , find the value of k to three decimal places.
Using $T_a = 20, T_0 = 90$, and $T(10) = 60$: $60 = 20 + (90 - 20)e^{-10k}$, so $40 = 70e^{-10k}$, giving $e^{-10k} = \frac{4}{7}$. Taking natural logs: $-10k = \ln \left(\frac{4}{7} \right) \approx -0.5596$, so $k \approx 0.056$.

- 18** This question has three parts. A tank initially contains 200 litres of pure water. Salt water containing 3 grams of salt per litre flows in at a rate of 2 litres per minute, and the well-mixed solution flows out at the same rate, so the volume stays constant at 200 litres. Let $S(t)$ be the amount of salt (in grams) in the tank at time t minutes. (a) Explain why $\frac{dS}{dt} = 6 - \frac{S}{100}$. (b) Solve this differential equation for $S(t)$, given $S(0) = 0$. (c) Find the amount of salt in the tank after 50 minutes, to the nearest gram.
- (a) Salt flows in at a rate of 3 grams/litre $\times$ 2 litres/minute = 6 grams/minute. Salt flows out at a rate proportional to the current concentration $\frac{S}{200}$ grams/litre, times the outflow rate of 2 litres/minute, giving $\frac{2S}{200} = \frac{S}{100}$ grams/minute out. Net rate: $\frac{dS}{dt} = 6 - \frac{S}{100}$. (b) This is a linear first-order DE. Integrating factor: $e^{\int \frac{1}{100} dt} = e^{t/100}$. $\frac{d}{dt}(Se^{t/100}) = 6e^{t/100}$. Integrating: $Se^{t/100} = 600e^{t/100} + C$, so $S = 600 + Ce^{-t/100}$. Using $S(0) = 0$: $0 = 600 + C$, so $C = -600$. Solution: $S(t) = 600(1 - e^{-t/100})$. (c) At $t = 50$: $S(50) = 600(1 - e^{-0.5}) = 600(1 - 0.6065) = 600(0.3935) \approx 236$ grams.